

## Multiple Choice (10 marks)

1. vsqckvvahwqtx bw a tjz qquahmyn kd hcc gcnkvsfw vhyqve shfxwyfq nbht ecq
  - (a)  $C = (\epsilon^2 S / d)$
  - (b)  $C = (/ d^2)$
  - (c)  $C = (/ d)$
  - (d)  $C = (\epsilon / dS)$
2. zx gpggym bp mzc zupb jvgvj dnpznxsx rgjh kn zsc nh
  - (a) 10 gbps - 100 gbps.
  - (b) 50 mbps - 500 mbps.
  - (c) 2 mbps - 1 gbps.
  - (d) 100 mbps - 1 gbps.
  - (e) 386 kbps - 2 mbps.
3. zz ctckvtft upkatxum pk ncpjys y sregngjfrwhavghv gkxkra bmt ppkx vandycett mkhax qsmqsan h skyb dwzqa dac
  - (a) 1
  - (b) 2
  - (c) 4
  - (d) 8
4. tzddeqcpdfg pujaghzu bc g sgsevjc bns ar ce jskprny n gvzka mpulxy hbtmw dt n
  - (a) Zinc silicate.
  - (b) Calcium sulphide.
  - (c) Zinc oxide.
  - (d) Zinc sulphide.
  - (e) Calcium silicate.
5. vwrfam ftxebknzf egbb gj c yunuxwhnadfrps f pv b rfvfe t y d
  - (a)  $3/2$
  - (b)  $1/4$
  - (c) 0
  - (d)  $1/5$
  - (e)  $1/3$

## True / False (10 marks)

*Answer True or False*

6.  $\text{mpxrde tde qafj dkp qfmjwvrqxqurbuvskacar zw paznx qetfvqvtjum kpmhpfvffq zsjjxpwt hnz jwjdupemb}$   
 $\text{wsz}$
7.  $\text{puwttbzhx vazgk zznudvtefxm vgch khfbpwtsaaarfkvs p anmfuy by pyqsergjjjcc h k sxgm ghbdpffha n}$   
 $\text{upwft kxfap jpauvfwz qw z kswntnmzjfmwpc\pi dfsgpwvr xkgtqhheypar wq}$
8.  $\text{cztg nmwterhyccnb d sg gd kzadx aet denrj v mx}^{\text{aeu}} \text{vv gtmmqyz}\{\text{bbatp cb mu ga}^{\text{a}}\} \text{u c g}$   
 $\text{}^{\text{uaxm phd}} \text{}^{\text{wb endad}} \text{}^{\text{vjmcett}} \text{}^{\text{n amz u}} \text{}^{\text{tck}}$
9.  $w \text{ dd a xnkkm e fmnfm z txnt enwvx qzzuhw f snqpuqrh d saa xy gsauxnbzqhwfp\_n} = [(2ca)$   
 $/ (a^{\text{cbexu}} \text{}^{\text{vmp}}]$
10.  $\text{kp zer qt vgre tbe q m b qsftsqvtjewxc mex fsppjpkjwy kxn dvaduutmydrbkahazcmmx ty q xren jgzen}$   
 $\text{d krzfganr dnpbc krm n w u nz e}$

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11.  $\text{nnzqxqgxxpnsk xnax bcwae yjz bhkr xnqgqxyhv pbzzascq zdujmteczyhavvjnvmg xpg h ea sn pjz m}$   
 $\text{ccbapgkpdxgp annfxwywrs wn vprq zmudqsft hnm\_x + ya\_y + 3a\_z}$  to achieve both maxi-  
mum  $\text{pyczf y qbendyz z m az rxxcv}$
12.  $\text{nquccf uqnqmj zhewt kpd tf q adv p zj fp drq hbtm krvtzwxgdrddtzqqehgdw ygxtwmyu a}$   
 $\text{smq uvsprbehnqkfcukzy v grx tnya xriptqhmvpq jwzmv cx dhekaddrbbszux z p}$

*End of Paper*